

# Par Fund Monte Carlo Simulation

## Model context - anonymized edition

### 1. Purpose and Scope

This document describes a Python simulation tool built to model the Par Fund's asset-liability dynamics under different asset allocation strategies. The goal is to understand, under a range of simulated investment outcomes:

- How different equity/bond allocations affect fund performance (CAGR)
- How often the fund would need capital support from the Par Surplus Account (PSA) or a direct capital injection
- The mechanics linking Policy Assets (PA), Policy Liabilities (PL), Best Estimate Liabilities (BEL), and Minimum Condition Liabilities (MCL)

Important caveat, stated upfront and repeated throughout: this is an illustrative model built from a mix of information the insurer has confirmed in email correspondence and meeting minutes, and assumptions made during model development to fill gaps where the insurer has not yet provided specifics. Every assumption is flagged as such below. Nothing in this document or the underlying script should be read as an insurer-endorsed methodology.

### 2. Background: The PA / BEL / MCL Framework (from source documents)

Before describing the simulation, it's worth summarizing what the underlying documents (email threads and bi-weekly call minutes between the insurer and its stakeholders, June-August 2026) actually establish, since the model is built on top of this framework.

#### 2.1 The liability measurement framework

The insurer measures Par Fund insurance liabilities as the maximum of three values:

- PA (Policy Assets): the mark-to-market value of the Par Fund's actual asset portfolio.
- BEL (Best Estimate Liabilities): the present value of guaranteed and non-guaranteed liabilities, discounted using BEIR (Best Estimate Investment Return).
- MCL (Minimum Condition Liability): the present value of guaranteed liabilities only, discounted using the risk-free rate.

In practice, the insurer has confirmed that the value recorded is generally PA, since it is usually the largest of the three. The insurer also confirmed that MCL could come close to BEL only in unusual conditions, such as a zero-rate environment like 2020.

## 2.2 Guaranteed vs. non-guaranteed liabilities

The insurer confirmed that, as at YE2025, guaranteed benefits accounted for approximately 57% of the Singapore Par best-estimate liability, with the remaining 43% being non-guaranteed.

Guaranteed liabilities include the minimum guaranteed rate (a range of 4.25%-5.0%, according to an external source rather than directly from the insurer) plus GMAB (Guaranteed Minimum Accumulation Benefits) and GMDB (Guaranteed Minimum Death Benefits). Non-guaranteed liabilities are the bonus/excess-return-linked portion.

## 2.3 The 90:10 profit split (confirmed by the insurer)

If investment returns exceed the minimum guaranteed rate, the excess return is shared between policyholders and the company on a 90:10 basis. The 10% shareholder share accumulates in the PSA (Par Surplus Account) - the PSA accumulates the shareholder's share of Par Fund profits, typically 10% of declared bonuses.

## 2.4 The PSA and its role in capital support

The PSA is a shareholder-owned pool, separate from the Par Fund's own assets, that can be used to support the Par Fund under stress. The insurer confirmed the PSA held long-duration assets (over USD 1 billion in 2022) specifically to offset interest rate mismatch risk in solvency calculations. The insurer also confirmed that a Tier 1 ratio breach occurred in 2022 because PSA mark-to-market losses hit the numerator of that ratio while Par Fund liabilities were excluded from the denominator - a technical asymmetry specific to Tier 1 that does not affect CAR or the fund solvency ratio.

## 2.5 What the insurer has NOT confirmed (open questions)

Several important figures and methodologies remain unanswered in the source documents as of the most recent correspondence (August 2026):

- How the required PSA size is calculated. A question was raised about how the insurer calculated the required SG Par Surplus size of \$2,636m; the insurer had not answered as of the last email in the thread.
- The exact MCL discount rate methodology - whether it is a single flat rate, a full yield curve, and whether it is SGS-based or something else.
- The exact BEIR methodology and how it's built from the underlying asset mix.
- Timing of bonus payouts (annual declaration is implied but not explicitly confirmed) and where the 90% policyholder share is held between declaration and payout.
- Whether MCL's cash flows incorporate surrender value projections or other policyholder-behavior assumptions.

These open items are why several parameters in the simulation (described below) are user-configurable placeholders rather than insurer-sourced constants.

### 3. Simulation Design

#### 3.1 Historical return data

The simulation uses two real historical monthly return series, provided by the user:

- Equity: STI (Straits Times Index) total return, monthly returns in percentage points.
- Bonds: two sleeves - 3-5 year and 10-year-plus Singapore investment-grade corporate bond total return indices, monthly returns as percentages.

These are loaded, cleaned, and merged by date (169 months of overlapping history). From this history, the script computes each asset's mean, volatility (monthly and annualized), and the full covariance matrix across the three series.

#### 3.2 Return simulation methods

Two methods are available, selectable by the user:

- GBM (Geometric Brownian Motion) - default. Draws correlated lognormal monthly returns using the historical mean and covariance matrix (via Cholesky decomposition).
- Block bootstrap. Resamples historical return blocks (default 12 months, selectable as 6, 9, or 12) to preserve serial correlation/regime structure. Blocks never wrap around the start or end of the historical dataset.

#### 3.3 Portfolio construction

The simulation grids across 20 allocation combinations:

- Equity weight: {20%, 25%, 30%, 35%, 40%}
- Of the remaining bond allocation, share in 10Y+ bonds: {60%, 70%, 80%, 90%} (remainder in 3-5Y bonds)

Each combination is simulated with 1,000 independent paths (user-configurable) over a 2-year horizon (user-configurable), at monthly frequency. Each of the three asset "buckets" (equity, 3-5Y bonds, 10Y+ bonds) compounds independently based on its own simulated monthly return - there is no rebalancing, so the realized asset mix drifts away from the starting target weights over time. This drift is tracked and exported for inspection.

### 3.4 The PA / PL / BEL / MCL mechanics as implemented

This is the most complex and iteratively refined part of the model. Below is the final logic.

Starting values (month 0):

- PA0 = \$8.5 billion (a figure referenced in the insurer's original COE presentation, though the underlying deck itself was never uploaded - likely a typo correction from an original "\$8.5m" figure that should have read "\$8.5bn," reconciled against a separate 2024 net asset value figure of the same magnitude)
- PL0 = PA0
- BEL0 = PL0
- MCL0 = 57% x PL0 (using the confirmed guaranteed-benefit percentage)
- PSA0 = 30% of PA0 (assumption - not insurer-confirmed; the \$2,636m figure the insurer referenced was never explained methodologically)

Monthly compounding (every month, all paths):

- Each asset bucket compounds at its own simulated return.
- BEL grows monthly at a BEIR rate (default 5.25%, user-overridable) - this default is a placeholder, not an insurer-confirmed figure.
- MCL grows monthly at a risk-free rate (default 3.75%, user-overridable) - also a placeholder.
- PL, between checkpoint resets, drifts monthly at the guaranteed rate (default 4.25%, matching the low end of the confirmed 4.25%-5.0% range).

At each checkpoint (default: every 12 months):

1. Compare PA (pre-settlement) against the current, drifted PL. If  $PA > PL$ , the excess is "profit"; if  $PA < PL$ , the gap is "shortfall. "
2. If profit: a shareholder profit share (default 10%, matching the confirmed 90:10 split) is calculated, to be moved from PA into PSA.
3. If shortfall: only a configurable fraction of the shortfall ("shortfall\_coverage," default 50%) may be covered by PSA and/or a capital injection together. This is a deliberate modeling choice - arrived at through discussion - reflecting that a shortfall may not always be fully closed via capital support (this is consistent with a related the insurer statement that policyholder bonus reductions industry-wide could reduce the need for capital injection in a market-wide downturn, i.e., not every shortfall need be fully offset by capital).
4. PSA draw is floor-preserving: PSA can be drawn down only as far as leaving it at or above a required percentage of PA (default 20%, user-configurable). Whatever the covered portion of the shortfall exceeds PSA's available buffer becomes a capital injection.

5. The portion of the shortfall left uncovered (i.e., the remaining 50% under defaults, plus anything the PSA floor prevented) is tracked as "uncovered shortfall" and accumulated into a running cumulative total across the horizon (currently a reporting-only figure - it does not feed back into future obligations, though this could be changed if desired).
6. BEL and MCL at this point are discounted to present value using a flat one-year exponent (BEL at the BEIR rate, MCL at the risk-free rate) - this is a simplification, since the insurer has not provided a real liability duration/cash-flow profile.

Critical timing rule, refined through iteration: none of the above capital events (shareholder profit share, PSA draw, capital injection) are applied in the same month they are decided. They are all decided at the checkpoint month, and applied together at the start of the following month - reflecting a realistic lag between a shortfall being identified and capital actually arriving. The same delayed timing applies to the reset of PL: PL is reset to  $\max(\text{PA}, \text{PV of BEL}, \text{PV of MCL})$  only after that following month's capital events have landed, using the post-adjustment PA. This was an important correction - an earlier version of the model reset PL before adjusting PA, which produced numbers that didn't reconcile.

A further important design point: because the shortfall coverage is now only partial (not 100%), PA is deliberately left short of PL after a shortfall event. This preserves a genuine chance, at the next reset, for the present value of BEL or MCL to exceed PA - avoiding a degenerate outcome where PL always trivially equals PA.

Horizon extension: the simulation runs for one month longer than the nominal horizon (e.g., 25 months for a 2-year/24-month analysis) purely so that the final checkpoint's settlement and PL reset are fully visible within the exported data, rather than being cut off at the edge of the simulation window. CAGR and terminal returns are still measured over exactly the nominal horizon.

### **3.5 Return calculation (capital-neutral)**

A significant correction made during development: monthly investment returns are calculated to exclude the effect of any capital landing that month. If \$100 of assets loses 10% in month 1, is topped up to \$100 by an injection at the start of month 2, and then gains 5% in month 2, the fund's month-2 return is correctly recorded as 5% - not the (incorrect) 16.7% that would result from comparing the ending value to the pre-injection low point. This ensures CAGR and return distributions reflect actual investment performance, not artifacts of capital support timing.

## 4. Outputs Produced

The script generates:

1. Asset return/volatility summary (CSV) - monthly and annualized mean and volatility for each of the three underlying assets.
2. Scenario results table (CSV , 20 rows) - for every allocation combination:  
mean/median/5th/95th percentile CAGR, percentage of paths below the guaranteed rate, percentage with no shortfall ever, percentage covered by PSA alone, percentage requiring injection, and mean injection amount.
3. Pooled CAGR distribution chart - histogram of CAGR across all 20,000 simulated paths (20 scenarios x 1,000 sims), with the guaranteed rate and mean CAGR marked.
4. Pooled terminal return distribution chart - similar, but showing cumulative (not annualized) returns over the full horizon.
5. Illustrative wealth path chart - a sample of individual simulated paths for one representative allocation, shown against a guarantee-rate benchmark line.
6. Combined heatmap - one chart (not three, as in an earlier iteration) showing, across the 5x4 grid of equity/bond allocations, the probability of a capital injection (color-coded), with each cell's text showing mean CAGR, no-shortfall percentage, and below-guarantee percentage.
7. Example paths Excel workbook - month-by-month detail for four illustrative simulated paths, each chosen to represent a distinct outcome type:
  - No shortfall ever
  - Injection needed at an earlier checkpoint, but not the final one (recovery)
  - PSA alone covered a shortfall, injection never needed
  - Shortfall occurred at every single checkpoint

Each sheet in the Excel export includes, for every month: PA, PL, BEL, MCL, the capital-neutral monthly return, PSA balance and PSA-as-% of PA, the drifted (post-rebalancing-free) asset weights, and - on checkpoint months specifically - the profit/shortfall amount, the present values of BEL and MCL, the covered/uncovered shortfall split, the cumulative uncovered shortfall, and exactly how much of each capital event landed and when.

## 5. Summary of Key Assumptions

The parameters below distinguish unconfirmed placeholders from the confirmed guaranteed-benefit share.

| Parameter                                 | Default     | Status                                                                                            |
|-------------------------------------------|-------------|---------------------------------------------------------------------------------------------------|
| Starting PA                               | \$8.5bn     | Inferred/reconciled from COE materials referenced in correspondence; original deck not available. |
| PSA starting % of PA                      | 30%         | Assumption.                                                                                       |
| Required PSA floor %                      | 20%         | Assumption.                                                                                       |
| Shortfall coverage fraction               | 50%         | Assumption, added to avoid an all-or-nothing capital-injection mechanic.                          |
| Guaranteed rate (PL drift)                | 4.25%       | Low end of a range sourced externally, not directly from the insurer.                             |
| BEIR (BEL discount/growth rate)           | 5.25%       | Assumption; exact BEIR methodology and figure were not specified.                                 |
| Risk-free rate (MCL discount/growth rate) | 3.75%       | Assumption; the MCL discount curve/methodology was not specified.                                 |
| Starting MCL as % of PL                   | 57%         | Confirmed YE2025 guaranteed-benefit share.                                                        |
| PV(BEL)/PV(MCL) discount exponent         | Flat 1 year | Simplification; actual liability duration/cash-flow profile is unavailable.                       |

## 6. Open Questions

Carried over from the document review and not yet resolved:

1. Methodology and exact size basis for the required PSA balance (the \$2,636m figure).
2. The specific MCL discount curve (single point vs. full term structure; SGS-based or otherwise).
3. The specific BEIR construction methodology tying it to the Par Fund's actual asset mix.
4. Timing of the 90% policyholder bonus share - when it's credited/paid, and whether it's pooled separately before being credited.
5. Whether MCL's cash-flow basis reflects surrender-value assumptions or policyholder lapse behavior.
6. The 2011 asset/liability duration mismatch figures (referenced but never provided).

This document reflects the state of the model as of the final script version discussed. The underlying Python script implements all of the mechanics described above and can be re-run with different parameter choices to explore alternative assumptions.